

PROJECT DESIGN PHASE-II

SOLUTION REQUIREMENTS

Functional & Non-Functional Requirements

Date	30 September 2026
Team ID	SWTID-2026-3024
Project Name	AI-Augmented Backend Development AI BlogNest API
Maximum Marks	4 marks

1. FUNCTIONAL REQUIREMENTS

The following are the functional requirements of the proposed AI BlogNest solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration through application form
		User login using registered credentials
		Password hashing using bcrypt.js
		JWT-based authentication
		Logout / session termination
FR-2	User Authorization & Role Management	Role-based access control
		Admin role management
		Editor role permissions
		Author role permissions
		Reader access permissions
FR-3	Blog Management	Restriction of unauthorized operations
		Create new blog posts
		Retrieve and view blog posts
		Update existing blog posts
		Delete blog posts
FR-4	Comment Management	Manage blog categories
		Manage blog tags
		Add comments to blog posts
		Retrieve comments
		Manage comments
FR-5	AI Blog Content Generation	Associate comments with relevant blog posts
		Accept topic or content-generation prompt

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Send prompt to configured AI service
		Generate AI-assisted blog content
		Return generated content through REST API
FR-6	AI Content Summarization	Accept existing blog content
		Process content using AI service
		Generate concise content summary
		Return summary through REST API
FR-7	Database Management	Store user information in MongoDB
		Store blog post information
		Store comments
		Store categories and tags
		Retrieve and update stored information using Mongoose
FR-8	API & Request Management	Provide RESTful API endpoints
		Process GET, POST, PUT/PATCH and DELETE requests
		Return structured JSON responses
		Handle invalid API requests
FR-9	Validation & Error Handling	Validate required input fields
		Reject invalid input data
		Handle authentication errors
		Handle authorization errors
		Handle database errors
		Provide meaningful error responses
FR-10	Security & Data Protection	JWT token verification
		Password hashing and protection
		Input sanitization
		Request validation
		Protection of protected API endpoints
FR-11	Search & Content Management	Search blog content
		Filter content using relevant categories or tags
		Retrieve matching blog records
FR-12	API Testing & Verification	Test authentication APIs
		Test blog management APIs
		Test AI generation APIs
		Test summarization APIs
		Test positive and negative scenarios using Postman

2. NON-FUNCTIONAL REQUIREMENTS

The following are the non-functional requirements of the proposed AI BlogNest solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The REST API should provide clear endpoints, structured responses, meaningful error messages, and consistent request and response formats so that developers and users can interact with the system easily.
NFR-2	Security	The system should protect user accounts and application data using JWT authentication, bcrypt.js password hashing, role-based authorization, input validation, and sanitization.
NFR-3	Reliability	The system should process valid requests consistently and handle invalid requests, database errors, authentication failures, and AI service errors without causing the application to crash.
NFR-4	Performance	API requests should be processed efficiently and responses should be returned within a reasonable time. Database queries and AI requests should be handled efficiently to support smooth system operation.
NFR-5	Availability	The backend system should be available whenever the server and required database and AI services are running. The system should handle service failures gracefully and provide appropriate error responses.
NFR-6	Scalability	The architecture should allow the system to support increasing numbers of users, blog posts, comments, API requests, and AI operations. Additional features and services should be integrable as the project grows.
NFR-7	Maintainability	The backend should use a modular structure with separate routes, controllers, models, configuration, and services so that the application can be maintained and extended easily.
NFR-8	Compatibility	The REST APIs should support standard HTTP requests and JSON-based responses so that they can be accessed by different client applications and API testing tools such as Postman.
NFR-9	Data Integrity	User, blog, comment, category, and tag information should be stored accurately and consistently in MongoDB. Validation should help prevent invalid or incomplete data from being stored.
NFR-10	Extensibility	The system should allow future integration of additional AI capabilities, content-management features, analytics, search improvements, and other services without requiring major changes to the complete architecture.

3. REQUIREMENT PRIORITY

The requirements can be categorized according to their importance to the core functionality of AI BlogNest.

High Priority

- User registration and authentication
- JWT authentication
- Role-based authorization
- Blog CRUD operations
- MongoDB database management
- AI blog content generation
- AI content summarization
- Security and validation
- Error handling

Medium Priority

- Comment management
- Categories and tags
- Search and filtering
- Advanced API validation
- Additional content-management features

Supporting Requirements

- Maintainability
- Scalability
- Compatibility
- Data integrity
- Extensibility
- Performance and availability

4. REQUIREMENT SUMMARY

AI BlogNest requires both functional and non-functional requirements to provide a complete AI-powered blog management backend.

The **functional requirements** define what the system should do, including authentication, authorization, blog management, comment management, AI content generation, AI summarization, database operations, API management, validation, security, and testing.

The **non-functional requirements** define the quality characteristics of the system, including usability, security, reliability, performance, availability, scalability, maintainability, compatibility, data integrity, and extensibility.

Together, these requirements provide the foundation for developing and evaluating the AI BlogNest system.

